

Academic Year	2025/2026
العام الدراسي	
Term	3
الفصل	
Subject	Chemistry /INSPIRE
المادة	الكيمياء / إنسباير
Grade	11
الصف	
Stream	Advanced
المسار	المتقدم
P 2	
Number of MCQ	30
عدد الأسئلة الموضوعية	
Marks of MCQ	من 2 إلى 4 درجات
درجة الأسئلة الموضوعية	
Number of FRQ	0
عدد أسئلة المطالبة	
Marks per FRQ	0
الدرجات للأسئلة المطالبة	
Type of All Questions	الأسئلة الموضوعية / MCQ
نوع كافة الأسئلة	
Maximum Overall Grade	100
Exam Duration	120 minutes
مدة الامتحان	
Mode of Implementation	SwiftAssess
طريقة التطبيق	
Calculator	Allowed
الآلة الحاسبة	مسموحة

Question*	Learning Outcome/Performance Criteria**	Reference(s) in the Student Book (English Version& Arabic Version)	
		المرجع في كتاب الطالب (النسخة الإنجليزية والنسخة العربية)	Page
السؤال*	نتائج التعلم / معايير الأداء**	Example/Exercise	مثال / تمرين
1	CHM.5.4.02.006.03 Calculate the equilibrium concentration given the value of the equilibrium constant and equilibrium concentration of reactants and products	Textbook + Example problem 4+ Practice Problems	96, 97
2	CHM.5.4.02.006.04 Calculate the value of K_{eq} , K_c , or K_p when the concentration or pressure of reactants and products are given at a constant temperature	Textbook + Example problem 4+ Practice Problems	96, 97
3	CHM.5.4.02.007.02 Write the solubility equilibrium expression for a given reaction equilibrium	Textbook	99
4	CHM.5.4.02.008.01 Calculate the molar solubility (concentration) of a sparingly soluble ionic compound using the solubility product constant, K_{sp}	Textbook + Table 3+ Example problem 5 + Practice Problems	100, 101
5	CHM.5.4.02.011.03 Explore the application of the solubility product constant in radiology while exploring its effects on human vital systems	Textbook + Figure 17	98, 99
6	CHM.5.4.02.007.03 Write the solubility for a given reaction equilibrium	Textbook + Table 3+ Example problem 5 + Practice Problems	100, 101
7	CHM.5.3.04.001.01 List six general properties of aqueous acids (taste, color of indicators, reaction with metals, metal carbonates and bases, and electrical conductivity)	Textbook + Figure 2+ Practice Problems	113, 114
8	CHM.5.3.04.001.03 Differentiate among acidic, basic and neutral solutions (in terms of the relative amounts of hydrogen ions and hydroxide ions)	Textbook + Figure 3	115
9	CHM.5.3.04.001.09 Describe the amphoteric behavior of water, H_2CO_3 , and ammonia, NH_3 (Using chemical equation, particulate diagram and space-filling models)	Textbook + Figure 7+ Practice Problems	116, 117, 118
10	CHM.5.3.04.001.10 Distinguish among monoprotic, diprotic and triprotic acids using ionization equations, examples and particulate diagrams	Textbook + Table 1	119, 120
11	CHM.5.3.04.001.08 Define acids and bases according to Brønsted-Lowry theory, indicating the acid, base, conjugate acid, conjugate base and conjugate acid-base pairs, when chemical equations, formula or space-filling models are given	Textbook + Figure 7+ Practice Problems	116, 117, 118
12	CHM.5.3.04.001.11 Define acids and bases according to Lewis theory	Textbook + Table 2	121
13	CHM.5.3.04.003.01 Compare between strong and weak acids (using examples, particulate diagrams and ionization equations)	Textbook + Table 3	124, 125, 126
14	CHM.5.3.04.006.01 Define acid ionization constant, K_a , while writing the ionization constant expression for different weak acids	Textbook + Table 4+ Practice Problems	126
15	CHM.5.3.04.003.02 Relate the strength of weak acids to the numerical values of K_a	Textbook + Table 4+ Practice Problems	126
16	CHM.5.3.04.003.04 Identify the relationship between the strength of an acid and its conjugate base and the strength of a base and its conjugate acid	Textbook + Practice Problems	125, 126
17	CHM.5.3.04.006.02 Define base ionization constant, K_b , while writing the ionization constant expression of different weak bases	Textbook + Tables 5, 6 + Practice Problems	127, 128
18	CHM.5.3.04.003.03 Compare between strong and weak bases (using examples, particulate diagrams and ionization equations)	Textbook + Tables 5, 6 + Practice Problems	127, 128
19	CHM.5.3.04.006.03 Relate the acidity and basicity of an aqueous solution to the hydronium and hydroxide ion concentration and pH at $25^\circ C$ or $298 K$	Textbook+ Figure 13 + Practice Problems	131, 132
20	CHM.5.3.04.007.01 Use K_w to calculate the hydronium ion and hydroxide ion concentration at a given temperature and vice versa	Textbook + Example problem 1 + Practice Problems	129, 130, 131
21	CHM.5.3.04.007.04 Calculate pOH when the $[H^+]$ or $[OH^-]$ is given and vice versa	Textbook + Example problems 2, 3 + Practice Problems	131, 132, 133
22	CHM.5.3.04.007.06 Calculate $[H^+]$ and $[OH^-]$ from pH	Textbook + Example problem 4 + Practice Problems	134
23	CHM.5.3.04.006.04 Calculate the acid dissociation constant, K_a , given acid concentration, $[H^+]$ and pH	Textbook + Example problem 5 + Practice problems	135, 136
24	CHM.5.3.04.005.07 List different methods used to measure the pH of a solution	Textbook + Figure 16	135, 136
25	CHM.5.3.04.004.01 Define neutralization reaction while writing the neutralization equation (Complete ionic and net ionic equations)	Textbook + Figure 18	138, 139
26	CHM.5.3.04.004.02 Define titration, and titrant CHM.5.3.04.004.03 Explain how to carry out an acid-base titration	Textbook + Figures 19, 23	139
27	CHM.5.3.04.004.04 Explain the difference between the equivalence point and the end point of titration process	Textbook + Figure 23	141, 142
28	CHM.5.3.04.009.02 Describe the titration curve of a weak acid with a strong base with respect to type of salt formed, pH and nature of solution at equivalence point, indicator used and its color change and volume of titrant needed for changing color of indicator	Textbook + Figures 20, 22	140, 141, 142
29	CHM.5.3.04.009.02 Describe the titration curve of a weak acid with a strong base with respect to type of salt formed, pH and nature of solution at equivalence point, indicator used and its color change and volume of titrant needed for changing color of indicator	Textbook + Figure 20	139, 140
30	CHM.5.3.04.004.06 Calculate the molarity (concentration) and volume of a solution using titration data	Textbook+ Problem solving strategy+ Example 6 + Practice Problems	142, 143
الأسئلة الموضوعية: MCQ			
+			
Questions might appear in a different order in the actual exam			
+			
قد تظهر الأسئلة بإرتيب مختلف في الامتحان الفعلي			
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As it appears in the textbook(UAE Edition INSPIRED Chemistry -Grade 11-12 (Volume 1) Advance Student Edition) , LMS, and (Main_IP).			
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